

A meta-analysis on effects of root development on soil hydraulic properties

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ABSTRACT

Root development can alter the structure of soil and its hydraulic properties, which in turn can affect runoff generation processes. Although some experimental studies have examined the relationships between root characteristics and soil hydraulic properties under certain conditions, the general patterns of the effects of root development on soil hydraulic properties under different conditions remain unclear. With an aim of examining the change patterns and factors influencing soil hydraulic properties induced by root development, we conducted a literature review and subsequent meta-analysis based on the evaluated literature data. Our findings revealed that root development significantly increased soil infiltration rates [average infiltration rate (AIR), 80.1%; initial infiltration rate (IIR), 81.8%; and steady infiltration rate (SIR), 144.9%], and saturated hydraulic conductivity (K_{sat}) by 46.3%. Among the selected indexes, only the non-capillary porosity showed moderate correlation with AIR and IIR ($R^2 = 0.6-0.65$), and root length density showed a low correlation with soil infiltration rates ($R^2 = 0.2-0.3$). In forestland, the increase in SIR was 5.6 times that of grassland; furthermore, SIR in soil associated with plants more than 5 years old was 3-fold higher than that observed in 1–5-year-old plants. Based on our analyses, we established that the identification of a general pattern in the relationship between root characteristics and soil hydraulic properties could not be achieved owing to the diverse influences of different conditions. For example, increases in soil infiltration rates and K_{sat} were found to be higher in loam soils and in soils with plants bearing a larger number of tap roots. Nevertheless, it can be implied that infiltration rates and K_{sat} gradually increase with plant age, and that root systems generally exert the most pronounced effect on infiltration rates in spring and summer. Accordingly, we recommend that future experimental research should focus on the effects of root systems under specific conditions, and that datasets based on observations recorded under multiple conditions should be compiled to establish general patterns based on the corresponding relationships under specific conditions. The information thus obtained would accordingly provide a basis for conducting further in-depth studies examining soil water movement and terrestrial hydrological cycle.

1. Introduction

Soil water movement is a vital process in the land phase of the hydrologic cycle, of which soil hydraulic properties are integral influencing factors. Soil hydraulic properties are dependent on multiple factors, including soil structure composition (e.g., particle size, porosity, organic matter content) (Ankenbauer and Loheide, 2017; Rawls et al., 2003; Russell, 2014; Stange and Horn, 2005; Yang et al., 2014), vegetation (Gajić et al., 2008; Wang et al., 2013), and human activities (cultivation and artificial additives) (Amoakwah et al., 2017; Bescansa et al., 2006; Kool et al., 2019; Obia et al., 2016). As prominent components of soil surface layers, the root systems of plants can have pronounced effects on the physical, chemical, and biological properties of

soil (Fageria and Stone, 2006), including changes in soil porosity, organic matter content, and the number and distribution of organisms and microorganisms, which can contribute to the modification of soil hydraulic properties (Hallett et al., 2003; Whalley et al., 2004). Accordingly, to gain a comprehensive understanding of soil water movement and the land phase of the hydrologic cycle, it is necessary to elucidate the mechanisms by which root systems affect soil hydraulic properties.

The findings of previous studies have revealed that the effects of root systems on soil hydraulic properties are mainly attributable to the processes of elongation, decay, secretion, and water-uptake (Benegas et al., 2014; Bengough, 2012; Jiang et al., 2018; Scholl et al., 2014; Young, 1998) (Fig. 1). During their elongation, roots penetrate soil pores and

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wrap around soil particles. Considering their different shapes and distribution characteristics, root systems can exert diverse effects on soil structure (Logsdon et al., 2013; Scholl et al., 2014; Zhang et al., 2019). Numerous studies have shown that roots within a specific range of diameters (i.e., 0.5 to 2 mm) exert significant effects on soil hydraulic properties, whereas comparably, roots with notably small or larger diameters may affect such properties to a less remarkable extent (Cui et al., 2019; Li et al., 2013; Liu et al., 2007). Furthermore, considering that herbaceous plants mainly possess thin shallow roots, and shrubs and trees possess larger taproots and deep root systems, root systems would be expected to exert different effects on soil hydraulic properties, and some can increase up to 5-fold (Jiang et al., 2018; Wang et al., 2018; Zhang et al., 2019). Moreover, it has also been found that the effects of root systems differ depending on the stage of plant growth. For example, during the early stage of growth, when root biomass is low, root system growth may promote soil drying–wetting cycles and the development of cracks within the soil, thereby increasing infiltration rates (Jotisankasa and Sirirattanachai, 2017). However, after the root biomass attains a certain threshold, the main effect of the root system on soil pores includes occupation, which may cause reductions in the density of soil pores and infiltration rates (Barley, 1954; Jotisankasa and Sirirattanachai, 2017; Leung et al., 2015a; Ni et al., 2019). In response to continued root growth, there are increases in soil porosity, and this results in augmented infiltration rates (Vergani and Graf, 2016). Additionally, coarse roots may form root channels within the soil and thereby promote rapid water movement, that is, the formation of preferential flow (Benegas et al., 2014; Cheng et al., 2017; Ghestem et al., 2011; Zhang et al., 2017). Root decay occurs throughout the growth process, and the rate of decay tends to differ according to the growth stage. When the extent of decay exceeds that of occupation, there is a concomitant increase in the soil infiltration rate (Ni et al., 2019), as root channels generated by root decay can increase the number of soil macropores (Guo et al., 2019; Meek et al., 1992; Mitchell et al., 1995; Ni et al., 2019; Zhang et al., 2020).

The effects of root exudates exerted on soil hydraulic properties are mainly manifested via three processes (Bengough, 2012; Gao et al., 2018; Yang et al., 2014). Root exudates can “glue” soil particles, thereby altering the number and size of soil aggregates and subsequently affecting soil infiltration. For example, a high concentration of mucilage may reduce the number of soil pores, thereby decreasing soil infiltration, whereas low concentrations of mucilage can moderate the diameter of soil aggregates and increase the structural stability of soil, which is conducive to water flow, thereby increasing soil infiltration (Janakiram et al., 2019). Furthermore, hydrophobic exudates can stabilize soil

structure and reduce changes in porosity, which decreases soil infiltration rates (Czarnes et al., 2000); however, this effect may be soil-type dependent (Naveed et al., 2018). Additionally, exudates can influence the types and distribution of rhizosphere-inhabiting organisms and microorganisms, which in turn can contribute to the modification of soil structure and hydraulic properties (Morales et al., 2010). However, these effects tend to be variable, considering the diverse range of exudate types and contents (Feeney et al., 2004).

Root water uptake, driven by plant transpiration, can affect soil structure by causing the occurrence of drying–wetting cycles, thereby affecting soil hydraulic properties (Czarnes et al., 2000; Gao et al., 2018; Lu et al., 2019). Moreover, such an uptake mechanism can affect soil hydraulic conductivity by directly influencing soil suction (Fatahi et al., 2010; Indraratna et al., 2006; Ng et al., 2013; Tötze et al., 2017). However, considering that changes in suction are dependent on multiple factors, including soil type, plant type, and climatic conditions (Leung, 2016), the degree of these effects can vary according to the prevailing conditions. The effects of root water uptake on soil suction may be more pronounced at low root densities, high water uptake rates and for dry soils (Williams, 1974). In addition, when plants grow to a certain stage, the influence of root water uptake on soil suction no longer increase (Ni et al., 2018). Moreover, compared with the changes in soil infiltration attributable to root elongation and decay, the contribution of root water uptake to changes in suction may be relatively less remarkable (Leung et al., 2015b).

Thus, it may seem that mechanisms underlying the effects of root development on soil hydraulic properties are particularly complex, differing to varying extents depending on the external conditions such as soil type, plant type, plant age, growth season, and soil depth (Archer et al., 2016; Hao et al., 2020; Jotisankasa and Sirirattanachai, 2017; Li et al., 1992; Liu et al., 2020; Ni et al., 2019). Presently, however, despite the availability of a substantial number of relevant studies, there is generally a lack of reliable quantitative data on the effects of roots on soil hydraulic properties. With an aim of systematically assessing the patterns whereby root systems influence soil hydraulic properties, we conducted a review of the existing literature and attempted to quantitatively analyze these patterns based on literature data. Based on our findings, we propose the following two hypotheses: (1) the root-induced changes in soil infiltration rates are attributed to changes in soil porosities and soil organic matter contents (Fig. 2); and (2) changes in root characteristics (including root biomass, root volume ratio, and root length density) can directly account for changes in soil infiltration rates. In this study, we performed a literature-based *meta*-analysis to integrate the available data from experimental results under different conditions

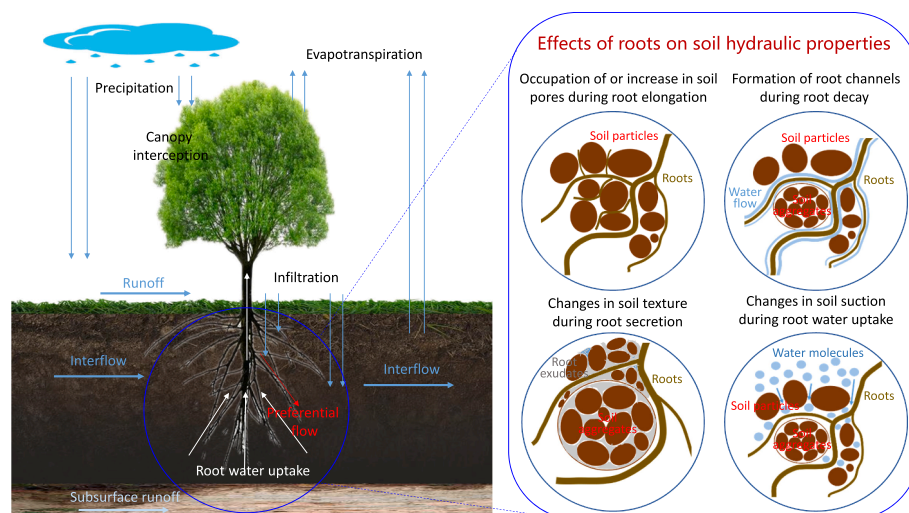


Fig. 1. The main mechanisms underlying the effects of root development exerted on soil hydraulic properties.

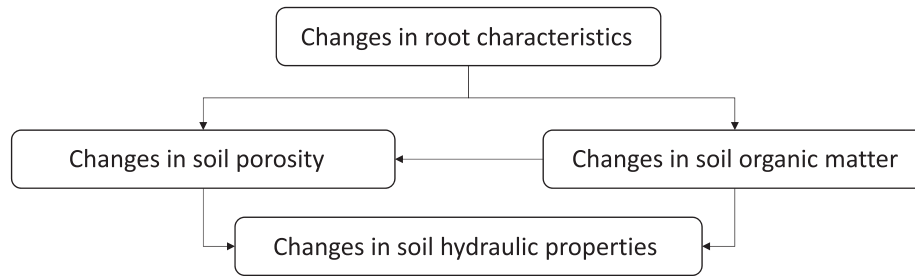


Fig. 2. The processes by which changes in root growth influence soil hydraulic properties.

and analyzed the root-induced changes in soil hydraulic properties. We then verified the correlation between these changes and soil porosity, soil organic matter content, and root characteristics. Moreover, we performed a comparison of the responses of soil hydraulic properties to root system changes under different soil and vegetation conditions. We anticipate that the findings of this analysis will provide a quantitative basis for the conduction of further in-depth studies on soil water movement and processes in the land phase of the hydrologic cycle.

2. Material and methods

2.1. Data collection

We used the Web of Science, Google Scholar, and China National Knowledge Infrastructure (CNKI) databases to explore related studies published before October 2020, Using the search terms “root” and “soil” and “infiltration” or “permeability” or “hydraulic properties”. We focused on the soil infiltration rates [average infiltration rate (AIR), initial infiltration rate (IIR), and steady infiltration rate (SIR)], soil saturated hydraulic conductivity (K_{sat}), soil porosity [total porosity (TP), capillary porosity (CP), and non-capillary porosity (NCP)], and soil organic matter content (SOM) in response to root system changes. Our selection of suitable publications was based on the following selection criteria: studies should (1) examine the influence of plant roots on soil hydraulic properties; (2) include laboratory or field experiment; (3) include at least one of the aforementioned four response variables of soil infiltration rate and saturated hydraulic conductivity; (4) include data from experimental group and control groups and (5) report sample sizes, mean values, and standard deviation (SD) or standard error (SE) of the data, or enable the calculation of SD/SE from the data provided (SE can be converted to SD using the formula $SD = SE\sqrt{n}$). Among the articles screened, 32 satisfying our stated criteria were eventually selected (Table A1), which included 508 experimental evaluations. Data were extracted from tables or figures using Getdata Graph Digitizer 2.20.

For each experimental assessment, we calculated and recorded the response ratio of the response variable, as well as information pertaining to the experimental site, measurement method, soil texture, plant type, plant age, growing season, and soil depth (Table A1). Soil texture was categorized into three groups (based on the China soil texture classification) (Deng, 1986), namely, sandy, loam, and clay soils. Plant types were categorized into five groups, namely grass, legumes, grass + shrub, shrub, and tree. Plant ages were categorized into three groups, namely ≤ 1 year, 1–5 years, and >5 years; growing seasons were categorized into four groups, namely spring, summer, autumn, and winter; and soil depths were categorized into five groups, namely total, 0–10 cm, 10–20 cm, 20–30 cm, and >30 cm. The seasons were categorized according to the “Standard for Division of Climatic Season (QX/T 152–2012)”, mainly using the pentad mean temperature method combined with the meteorological division method. According to historical records, in the temperate regions of the northern hemisphere with four distinct seasons, spring is generally from March to May, summer is from June to August, autumn is from September to November, and winter is from December to February of the following year. In other regions, the experimental season

was inferred based on the information of experimental location and time provided by each record, combined with the local historical temperature data.

2.2. Data analysis

In this study, the *meta-analysis* approach was used to analyze the effects of root systems on soil hydraulic properties. Statistical analysis was performed by Stata 12.0 based on a random-effects model. When analyzing the changes in response variables, the natural logarithm of the response ratio (R) was used as the effect size of each study according to the method presented by Hedges et al. (1999) and was recorded as RR in this study. Calculation of the effect size is shown in Eq. (1), variance (v) of the effect size is calculated by Eq. (2), and the 95% confidence interval (CI) is calculated by Eq. (3).

$$RR = \ln(R) = \ln\left(\frac{X_t}{X_c}\right) \quad (1)$$

$$v = \frac{S_t^2}{n_t X_t^2} + \frac{S_c^2}{n_c X_c^2} \quad (2)$$

$$95\%CI = RR \pm Z_{\alpha/2} \sqrt{v} \quad (3)$$

where X_t , S_t and n_t denote the mean, standard deviation and sample size of the experimental group, respectively, and X_c , S_c and n_c denote the mean, standard deviation and sample size of the control group, respectively. The value of $Z_{\alpha/2}$ is 1.96 ($\alpha = 0.05$).

In the random-effects model, the weighting factor (w) of RR in each study was the reciprocal of the total variance (Eq. (4)), the weighted effect size (RR^*) was calculated by Eq. (5), and the standard error SE (RR^*) was calculated by Eq. (6) (Hedges et al., 1999).

$$w = \frac{1}{v + \tau^2} \quad (4)$$

$$RR^* = \frac{\sum_{i=1}^k w_i RR_i}{\sum_{i=1}^k w_i} \quad (5)$$

$$SE(RR^*) = \sqrt{\frac{1}{\sum_{i=1}^k w_i}} \quad (6)$$

where v denotes the within-study random variance and τ^2 denotes the between-study variance.

If the 95% confidence interval (CI) did not overlap with zero, the effects of the root system on the response variables were considered to be statistically significant. A value greater than zero indicated a positive impact, and a value less than zero showed a negative impact. Heterogeneity between groups (Q) was used to test the differences in weighted effect sizes between each group. If $p < 0.05$, this means that the difference is significant.

The change rate (CR) of the effect size was calculated by Eq. (7).

$$CR = (e^{RR} - 1)\% \quad (7)$$

3. Results

3.1. Overall changes in response variables

By calculating the weighted average effect size, we found that the infiltration rates and soil saturated hydraulic conductivity showed significant increases under the influence of root system development (Fig. 3), for which SIR showed the highest increase, at 144.9%. K_{sat} increased by 46.3% and AIR and IIR increased by 80.1% and 81.8%, respectively. We speculated that these increases in infiltration rates and saturated hydraulic conductivity might be attributable to the changes in soil porosity and soil organic matter (SOM) content, and consequently, we also analyzed changes in soil porosity and SOM content under the influence of root system development. The results obtained (Fig. 3) revealed that soil porosity and SOM content also showed increasing trends. The three types of soil porosity (TP, CP, and NCP) were found to show relatively small increases of 7.3%, 7%, and 15%, respectively. In contrast, the increase in SOM content was relatively considerable at 51.3%.

3.2. Changes in response variables under different conditions

We found that the aforementioned eight response variables showed different patterns of change for different soil textures, plant types, plant ages, growing seasons, and soil depths. For example, as depicted in Fig. 4a and Table A2, the changes in AIR differed notably among the three assessed soil textures, with values of only 18.6% in sandy soil, and 101.6% and 74% in loam and clay soils, respectively. Similarly, we observed a wide variation in the changes in AIR with respect to different plant ages, with the most remarkable increase of 127.8% occurring in soils associated with 1–5-year-old plants, whereas in the case of plants with an age <1 year, there was a less remarkable increase of 18.6%. When grouped according to the plant type and growing season, the heterogeneity between groups was found to be small, whereas with regards to different soil depths, the increases among different soil layers (131.4%–169.2%) seemed to be greater than that of the whole soil profile (62.2%), although the confidence intervals were correspondingly greater.

Furthermore, with respect to soil texture, it was found that increases in IIR in loam and clay soils reached 94.8% and 91%, respectively, whereas no significant changes were observed in sandy soil (Fig. 4b, Table A2). Additionally, increases in IIR were found to be notably higher in forestland, reaching 159.4%, compared with the small differences in soils under grass, legumes, and shrubs (28.5%, 54.7%, and 66.4%,

respectively). Moreover, increases in IIR in soils associated with plants aged 1 to 5 years and in soils associated with plants aged more than 5 years reached 96.3% and 150.6%, respectively, whereas the changes for plants aged <1 year were not significant. In general, the increases in IIR in different growing seasons did not differ substantially (81.7%–100.8%), and similar to the response of AIR, increases in different soil layers were found to be greater than that for overall soil depth.

SIR exhibited significant heterogeneity in each of the subgroup analyzed (Fig. 4c, Table A2). Among the three soil textures, SIR showed the most remarkable increases in loam and sandy soils at 155.2% and 121.9%, respectively, whereas for clay soils, the increase was only 12.2%. With respect to the vegetation type, the increase in SIR in soils under grass was relatively less remarkable (61%), compared with that under trees (340.4%), which represented a 5.6-fold difference. However, there were no significant changes in SIR in soils under legumes. In terms of plant age, SIR increased with increasing age, with increases of 131.1% and 355% reported for plants aged 1 to 5 years and for plants aged more than 5 years, respectively, which represented an approximate 3-fold difference. The changes in SIR were also found to be affected by season, with rates of increase reaching 113.1% and 194.8% in spring and summer, respectively, whereas rates were notably lower at 42.6% in autumn. With respect to different soil depths, the response of SIR was found to be similar to that of AIR and IIR, with considerable increases in different soil layers compared to those in the whole soil profile.

Analyses of the changes in saturated hydraulic conductivity (Fig. 4d, Table A2) in each subgroup revealed that most changes were non-significant. With respect to the soil texture, however, we found that the increase in loam soils reached 106.7%, whereas no significant change was detected in sandy soils. Similarly, among the different vegetation types, K_{sat} only showed significant changes in soil under trees, and in plants greater than 5 years of age among the plants belonging to different age groups, with increases of 191.7% and 146%, respectively. From the perspective of growing season, although the changes in summer and winter were not significant, there were significant increases detected in spring and autumn, even though considering the relatively small sample size, the results might reflect random effects. At different soil depths, the response pattern of K_{sat} was found to be similar to that observed with the three infiltration rates. However, owing to the small sample size for different soils layers, the results may also be attributable to random effects.

With regards to porosity, the changes in TP and CP under different conditions tended to be relatively less remarkable (Fig. 4e, f, Table A2). Increases in the TP of loam and clay soils were found to be similar at 7.6% and 7%, respectively, although for CP, the increase in loam (13.1%) was higher than that in clay (1.3%). Furthermore, compared with other plant types, the increases in TP and CP (11.5% and 10.8%, respectively) were both substantial in soil under trees, whereas with increasing plant age, there were slight increases in both TP and CP. Seasonally, we detected slight differences in the increases in TP, although increases in CP tended to be slightly greater in summer than those observed in autumn. Consistently, differences in the changes in TP and CP among the different soil layer were relatively less remarkable.

It was difficult to discern any clear patterns in the changes in the NCP under different conditions (Fig. 4g, Table A2). For example, with respect to soil texture, although NCP showed a significant increase in clay soil (40.2%), no significant change was detected for loam soil. Similarly, changes in the NCP in soil under grass and trees were non-significant, as were those in soils associated with plants aged <1 year and more than 5 years, and those in spring and summer.

Differences in the increase in SOM for each condition were relatively less remarkable, with values ranging from 32% to 85% (Fig. 4h, Table A2). Among the three soil textures, the increase in SOM content in clay soil (85%) was notably greater than that in loam soil (37%). From the perspective of plant type, the most pronounced increase in SOM was in the soil under shrubs at 80.7%, whereas for other plant types, increases of approximately 40% were reported. Moreover, the increases in

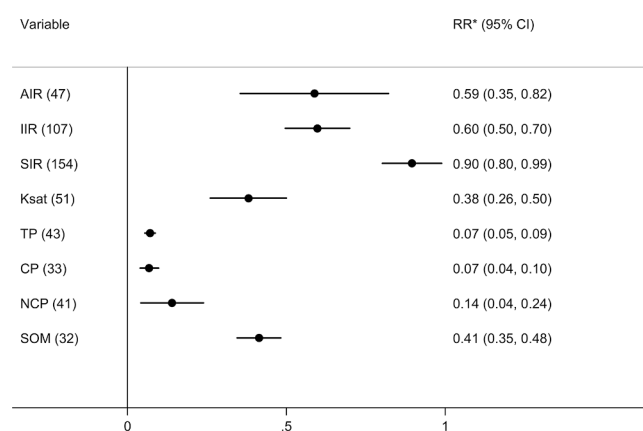


Fig. 3. Changes in response variables. Data listed on the right-hand side show weighted effect sizes (RR*) and their 95% confidence intervals. The number of studies are shown in parentheses on the left-hand side.

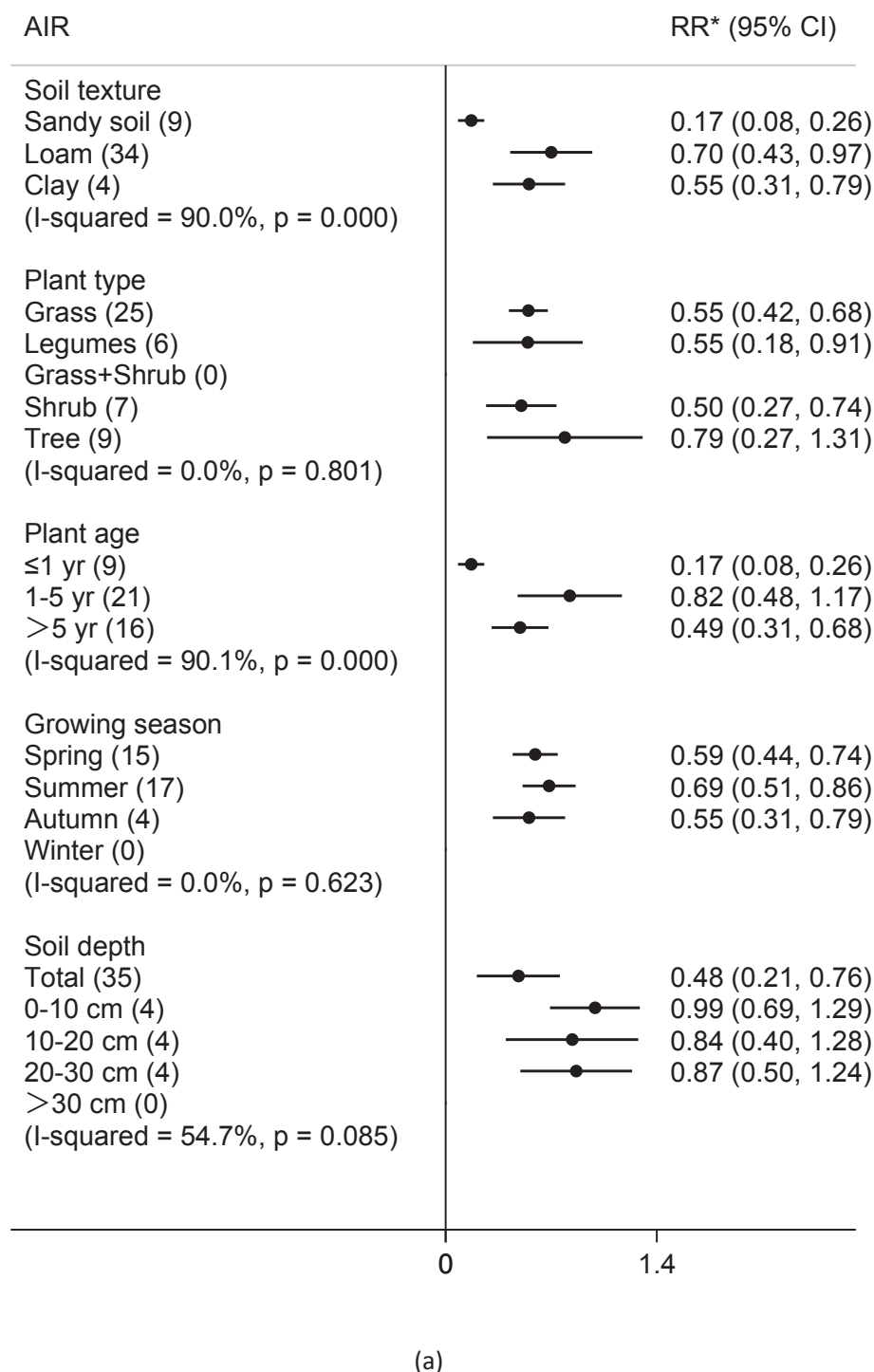


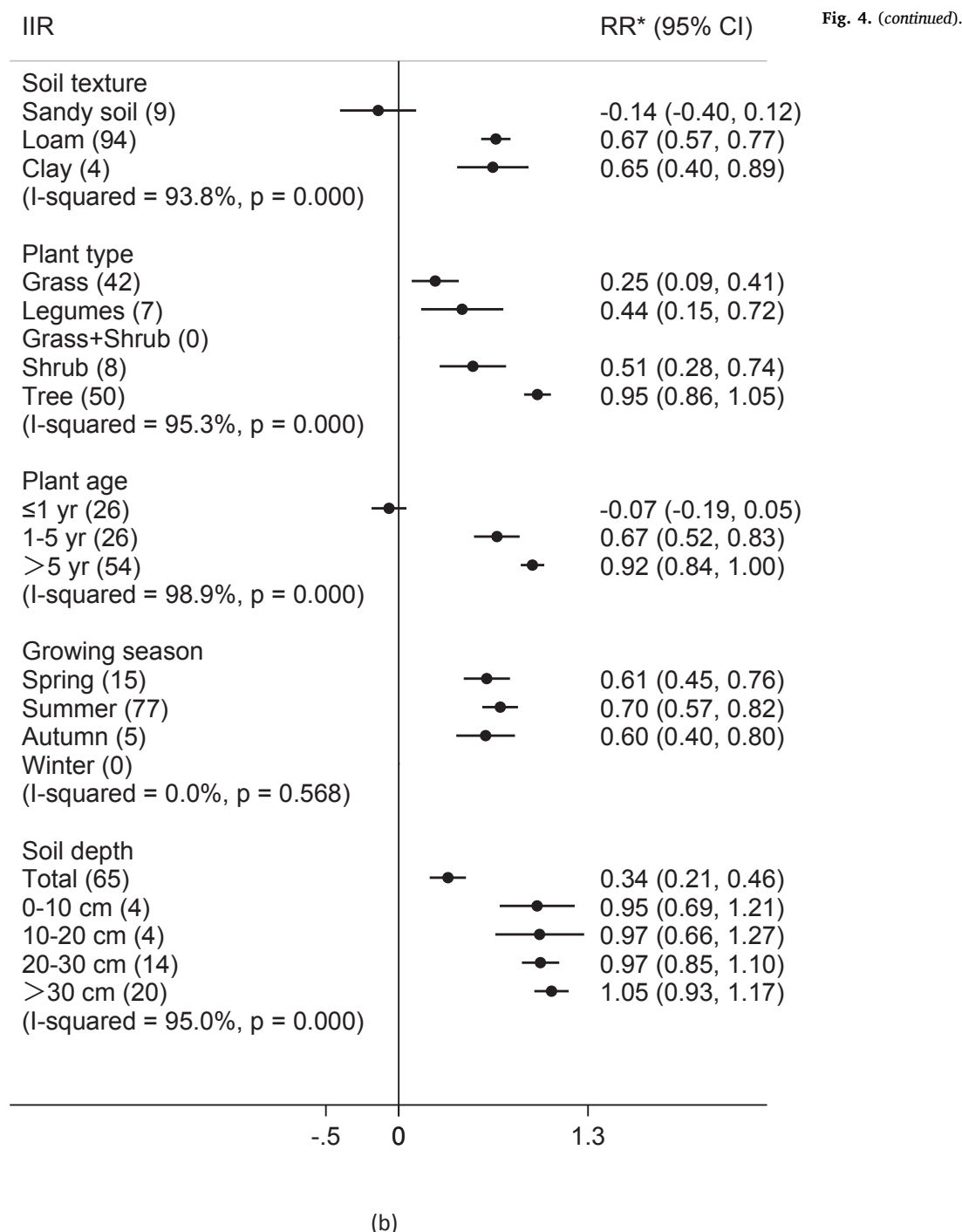
Fig. 4. Changes in average infiltration rate (AIR) (a), initial infiltration rate (IIR) (b), steady infiltration rate (SIR) (c), saturated hydraulic conductivity (K_{sat}) (d), total porosity (TP) (e), capillary porosity (CP) (f), non-capillary porosity (NCP) (g), and soil organic matter content (SOM) (h) under different conditions. The groups of the five conditions are shown on the left-hand side and the sample sizes are shown in parentheses. I^2 denotes the degree of heterogeneity between groups, P denotes credibility, and a P -value < 0.05 represents apparent heterogeneity. On the right-hand side, the RR^* (weighted effect size) values with 95% confidence intervals have been indicated.

SOM in autumn (85%) were found to be greater than those observed in the other seasons. However, when grouped by plant age and soil depth, there seemed to be less heterogeneity in the changes of SOM among the assessed groups.

3.3. Correlations among changes in soil porosity, soil organic matter, and soil infiltration rates

Although linear correlation analyses performed to assess associations between the changes in soil porosity and those in infiltration rates

yielded P values of <0.05, the corresponding correlation coefficients (R^2) tended to be low (Fig. 5). Changes in TP were significantly and positively correlated with changes in the three assessed infiltration rates ($P < 0.05$), whereas R^2 values did not exceed 0.2. Similarly, changes in CP were significantly positively correlated with changes in AIR and SIR ($P < 0.001$) and significantly negatively correlated with changes in IIR ($P < 0.001$) with R^2 values of <0.1. Changes in NCP were significantly and positively correlated with changes in all three infiltration rates ($P < 0.01$), whereas the correlation coefficients for AIR and IIR were relatively greater and approximated to a linear correlation. Changes in SOM



contents have no significant correlation with the changes in the three infiltration characteristics and the three porosities ($R^2 < 0.1$) (Fig. 6).

3.4. Correlations between root characteristic variables and changes in soil infiltration rates

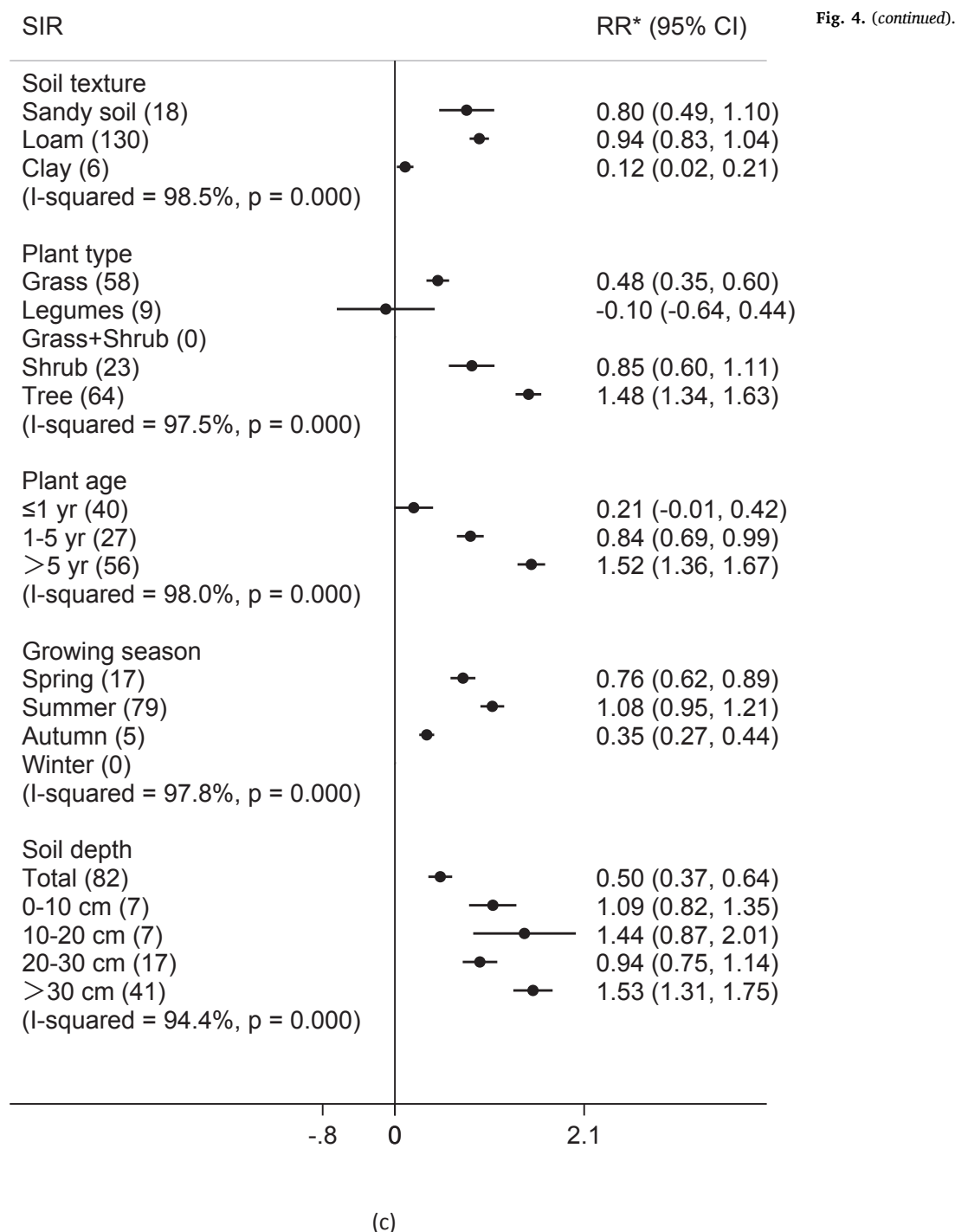
Linear correlation analysis performed to assess associations between five root characteristics and changes in soil infiltration rates revealed that root biomass (RB), root volume ratio (RVR), root length density (RLD), and root weight density (RWD) were significantly correlated with the changes in infiltration rates ($P < 0.05$), while the corresponding correlation coefficients exhibited small values ($R^2 < 0.3$) (Fig. 7). Correlation coefficients between RB and RVR and the changes in three

infiltration rates were all less than or equal to 0.1, indicating that there was no significant correlation. Correlation coefficient between RLD and the changes in three infiltration rates is slightly higher ($R^2 = 0.2-0.3$), but this may be due to individual data affecting overall trend, which is also reflect in RWD. All correlations between root surface area density (RSAD) and changes in the three infiltration rates were found to be non-significant (P greater than 0.05).

4. Discussion

4.1. Factors affecting soil infiltration rates

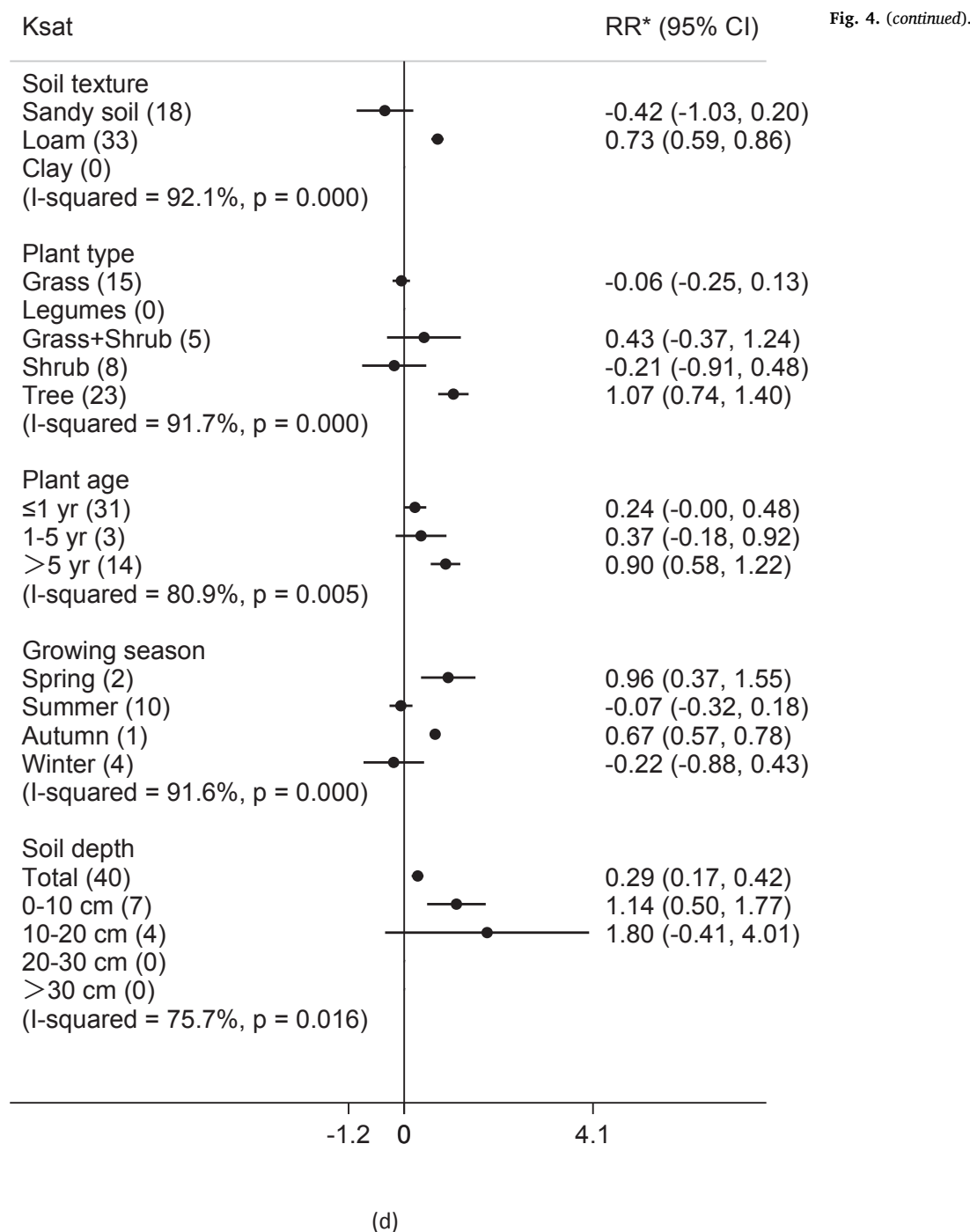
The findings of our meta-analysis (Fig. 3) indicated that the



infiltration rates, saturated hydraulic conductivity, porosity, and organic matter content of soils increased markedly in response to the influence of root development. In general, when soil reaches steady infiltration conditions, the water potential gradient is approximately 1, and the saturated hydraulic conductivity (K_{sat}) and steady infiltration rate (SIR) are similar (Shao et al., 2006). However, our findings indicate that there are significant differences between the changes in SIR and K_{sat} , which may conceivably be associated with the randomness of the samples. For example, most studies conducted on soil K_{sat} have been performed using plants aged <1 year, whereas a majority of studies reported on SIR have been performed using older plants. Moreover, whereas most studies conducted on SIR involve field experiments, those

conducted on K_{sat} are mostly laboratory-based.

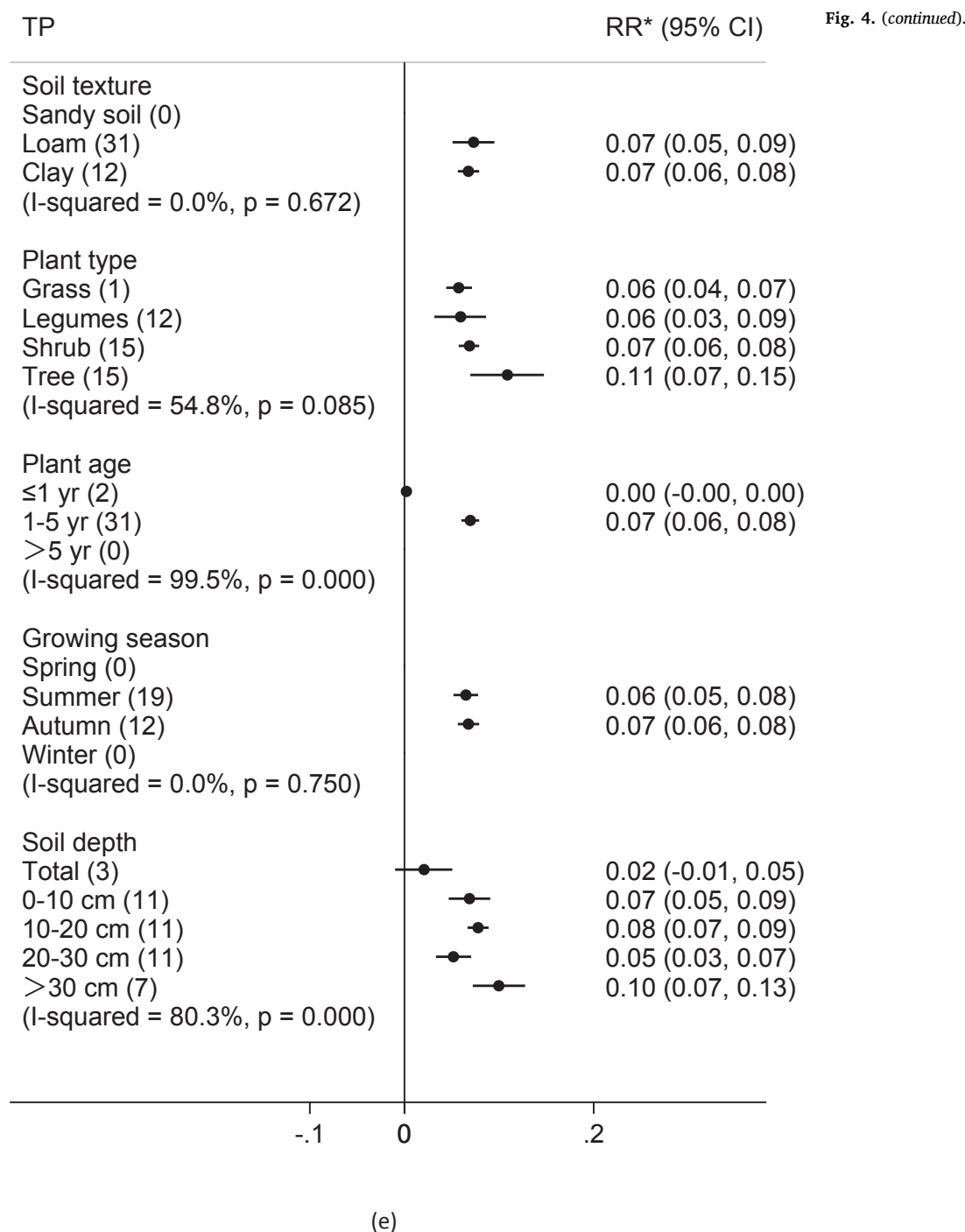
Based on our initial assumption that the effects of root development on soil infiltration rates would be reflected primarily in the changes in soil porosity and organic matter content, we initially assessed correlations between changes in soil porosity and those pertaining to soil infiltration rates (Fig. 5). The results accordingly revealed that correlations between the changes in total porosity (TP) and CP and changes in the three infiltration rates were less remarkable, while changes in non-capillary porosity (NCP) were found to exhibit relatively high correlation with changes in infiltration rates. These findings thus seem to indicate that non-capillary pores play a more important role in the occurrence of changes in infiltration rates (Mitchell et al., 1995;



Rachman et al., 2004; Rasse et al., 2000; Tang et al., 2019), particularly with respect to AIR and IIR. With respect to SIR, it is conceivable that patterns may be differentially affected by non-capillary pores under different conditions. For example, in a coffee plantation in Central America, the volcanic soil exhibits high natural permeability, and the effects of large pores created by root development on soil infiltration are not remarkable, whereas pores of a similar dimension have been found to significantly enhance soil permeability in pasture lands (Benegas et al., 2014).

Soil organic matter (SOM) mainly affects soil hydraulic properties by altering the structural and adsorption properties of soils (Gao et al., 2018; Saxton and Rawls, 2006; Yang et al., 2014). The analyses revealed that the correlation coefficient (R^2) values between the changes in SOM

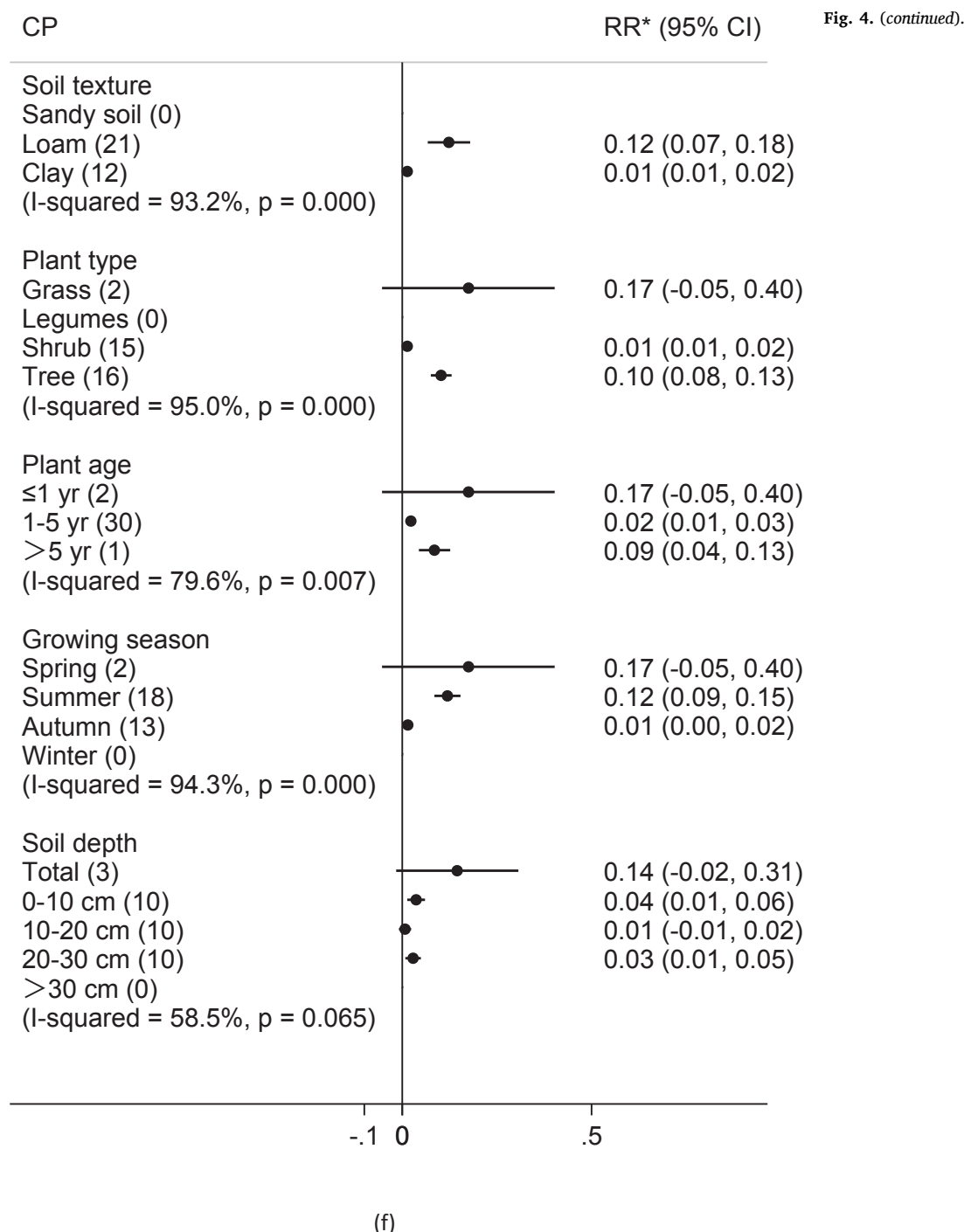
and soil infiltration rates and soil porosity were small and found to be <0.1 (Fig. 6). Based on these results, it is therefore difficult to identify any consistent pattern with regards to the effects of changes in SOM content on soil porosity or infiltration rates. Studies conducted on the effect of SOM have suggested that the concentration of soil organic contents (such as mucilage) secreted by roots will exert different effects on soil permeability. For example, at high concentrations, SOM can glue soil particles and thus alter soil pore dimensions, thereby decreasing water infiltration. Conversely, low SOM concentrations are favorable for an increase in the number of medium-sized soil pores and a relatively stable soil structure, which is in turn conducive to the flow of water and may contribute to an increase in water infiltration (Janakiram et al., 2019). Other studies have shown that the organic matter produced by



root systems may be hydrophobic or hydrophilic, and depending on soil moisture content, this will also accordingly promote a reduction or increase in soil infiltration rates (Bengough, 2012). Additionally, although the organic matter produced by root systems and rhizosphere microorganisms can enhance the strength of the bond formed between soil particles and reduce soil infiltration rates, reductions in soil hydrophobicity may occur as a consequence of multiple drying–wetting cycles, which may in turn increase soil infiltration rates (Czarnes et al., 2000).

Correlation analyses of the changes in root characteristics and soil infiltration rates (Fig. 7) indicated that although most of the assessed root characteristics were significantly correlated with changes in soil infiltration rates ($P < 0.05$), only root length density (RLD) was found to exhibit a low correlation with the three infiltration rates, whereas the

other characteristics showed no correlation ($R^2 \leq 0.1$). We speculate that these findings may be explained in terms of considerable differences in the degree of influence of root characteristics exerted on the changes in soil infiltration rate under different conditions. For example, Liu et al. (2020) have reported that RLD affects soil infiltration rates only at depths between 5 and 30 cm, whereas at depths ranging between 10 and 20 cm, the root surface area demonstrates a more pronounced influence. Moreover, the same characteristics of roots with different architectures may also exert significantly different effects on soil infiltration rates (Wang et al., 2018). Additionally, this may be associated with the fact that root characteristics are highly correlated with soil hydraulic properties within a certain threshold. For example, Vergani et al. (2015) have reported that soil permeability increases at RLD values of $<0.1 \text{ cm/cm}^3$,



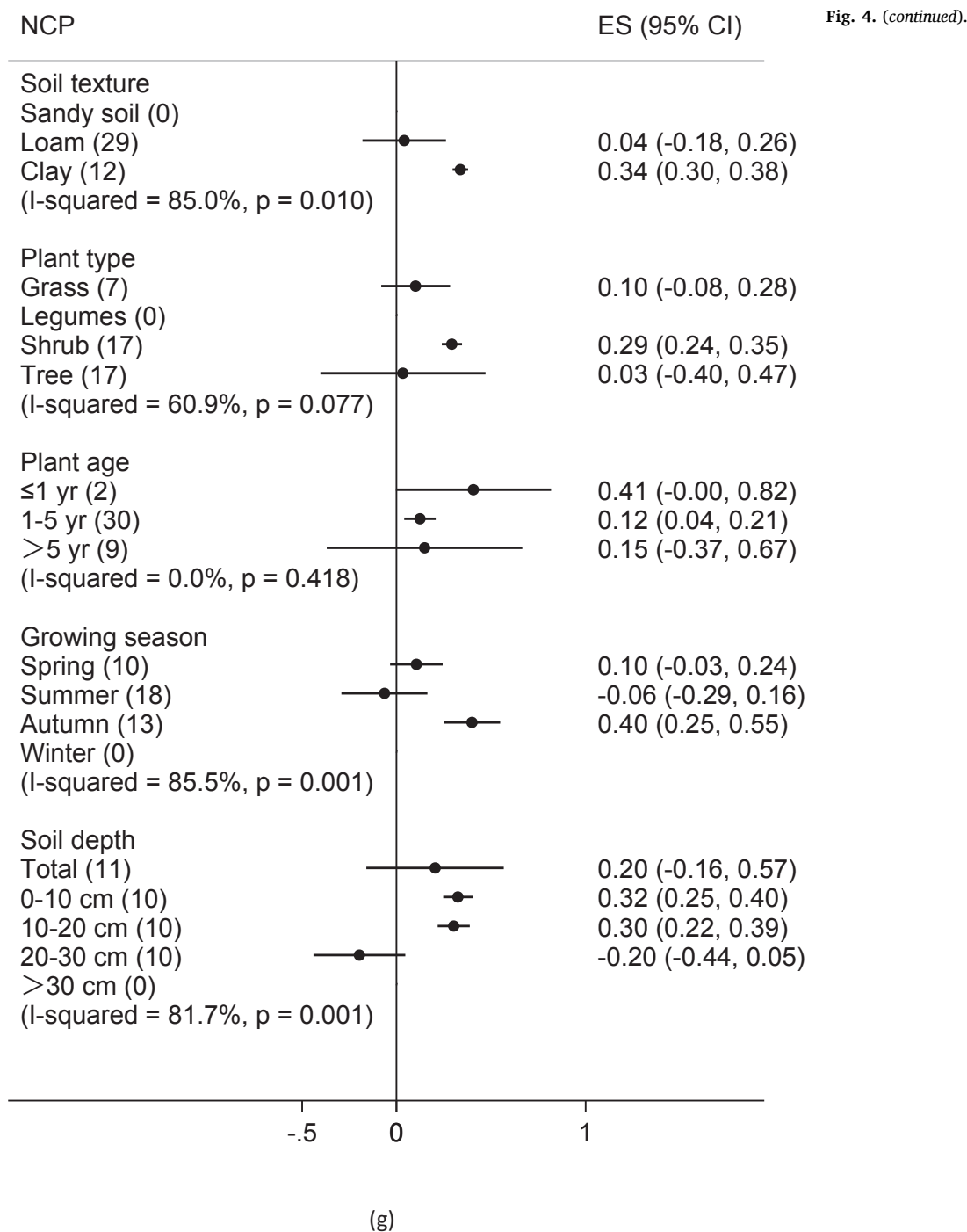
whereas beyond this threshold, there is a reduction in permeability (Vergani and Graf, 2016). Similarly, other studies have reported that the enhancement of soil permeability by root development is mainly attributed to the length and surface area densities of roots with diameters ranging between 0.5 and 5 mm (Li et al., 2013; Yan et al., 2016).

The findings of the aforementioned analyses thus indicate that when assessed under different conditions, changes in root characteristics, soil porosity, and SOM content show almost no correlation with the changes in soil infiltration rates, with only NCP presenting a moderate correlation with AIR and IIR, and RLD showing less correlations with the three infiltration rates. These observations are accordingly at odds with our stated hypotheses, which we suspect may be attributable to the fact that the extent of the influence of these soil properties differs according to the

soil and vegetation conditions. Upon assessing this possibility, we thus analyzed the effects of root systems under a series of different conditions.

4.2. The effects of root systems on soil hydraulic properties under different conditions

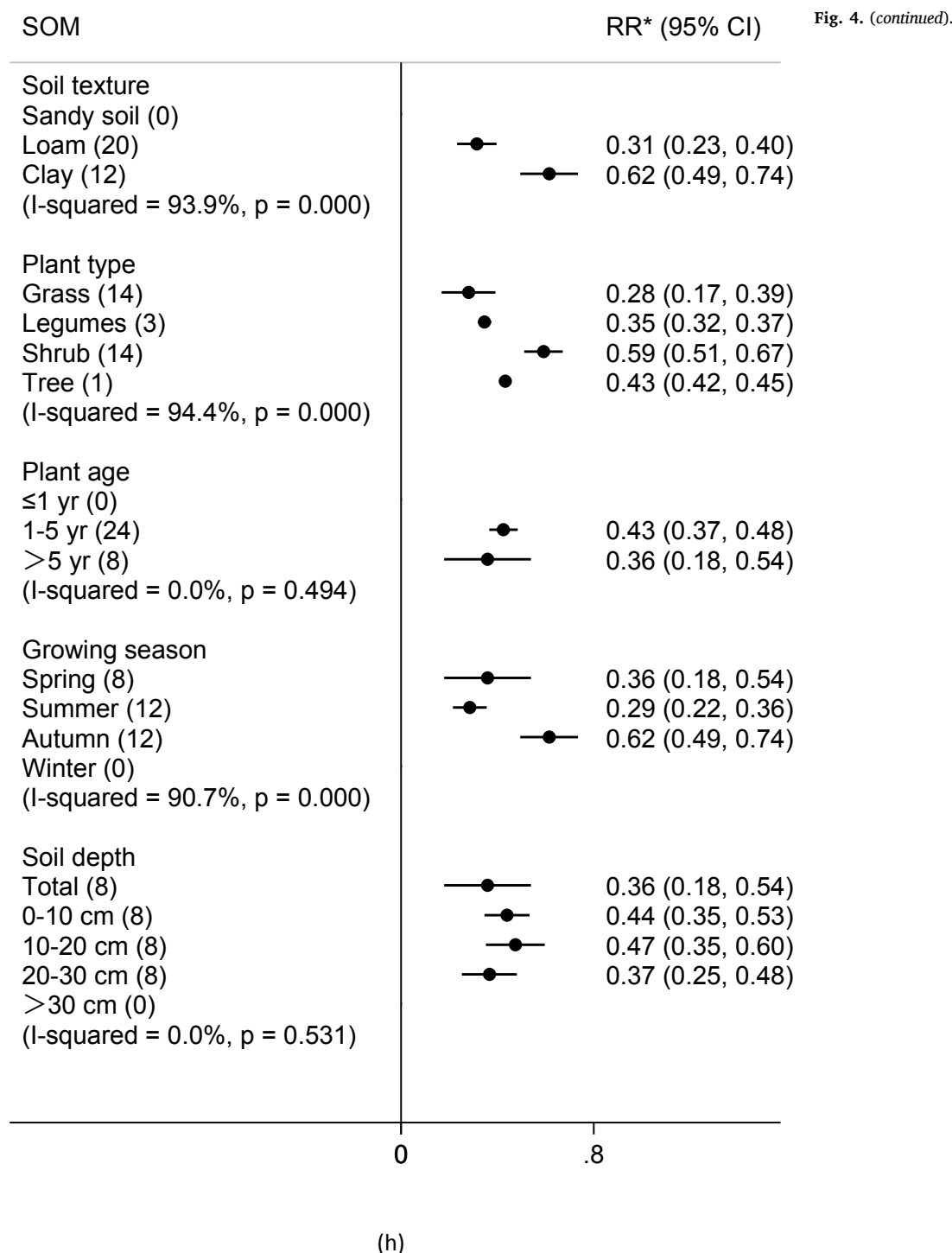
The results of our classification analyses (Fig. 4a, b, c, d) indicated that the influence of root system development on soil infiltration rates and saturated hydraulic conductivity differed according to the soil texture, plant type, plant age, growing season, and soil depth. Overall, we found that observed changes in SIR and K_{sat} were markedly affected by different conditions, whereas changes in AIR tended to be affected to



a lesser extent by the plant type and growing season, and that the growing season seemed to exert a less remarkable influence on the changes in IIR.

When considering the effects of soil texture, we found that increases in the three infiltration rates and saturated hydraulic conductivity were the most pronounced in loam soil, whereas changes to a differing extent were observed for sandy soil. The difference between them was conceivably associated with the compact soil grain structure and a relatively higher density of medium-sized pores in loam soil, and therefore root systems are more likely to increase porosity during their development, which is consistent with the increase in TP in loam soil shown in Fig. 4e. In contrast, the structure of sandy soil is notably less compact, with a greater number of macropores, and thus there is a

greater likelihood of pore occupation due to root development. Accordingly, whereas the pores in certain regions of soil may be occupied, the porosity of other regions may increase, and although pores in the same region may be occupied during the initial stages of plant growth, porosity may increase with plant growth (Leung et al., 2018; Ni et al., 2019; Scholl et al., 2014). Consequently, the overall changes in infiltration rates and saturated hydraulic conductivity may be non-significant. Increases in the infiltration rates of clay soil were found to be slightly lower than those in loam soil, which might be presumably attributed to the fine texture and numerous small pores that characterize clay soil. Moreover, a few pores generated by root growth (particularly that by fine roots) may be subjected to lateral pressure from the soil, thereby reducing the water conduction capacity of these pores, resulting



in reduced infiltration rates, whereas for plants with taproots, root development can continue to produce stable macropores to promote infiltration (Mitchell et al., 1995). These properties are consistent with those reported in the results shown in Fig. 4e, f, and g.

With respect to vegetation type, we found that the changes in infiltration rates and saturated hydraulic conductivity in soil under different plant types mainly showed a pattern of increase associated with the number of taproots, which is presumably indicative of an increase in porosity in soils containing a greater number of taproots (Fig. 4e, f, g), and thereby a larger network of water flow paths.

We found that with increasing plant age, infiltration rates and

saturated hydraulic conductivity showed increasing trends, which might be related to the fact that during the early stages of plant growth, occupation of soil pores due to root development might reduce infiltration rates (Barley, 1954; Leung et al., 2015a; Ni et al., 2019). As plants continue to grow, porosity increases, and root channels are generated during the process of root turnover, thereby contributing to a continual increase in infiltration rates (Guo et al., 2019; Meek et al., 1992; Mitchell et al., 1995).

In terms of seasonal effects, we found that increases in SIR tended to be the most pronounced during summer, followed by spring and autumn, which presumably reflected the seasonality of root system

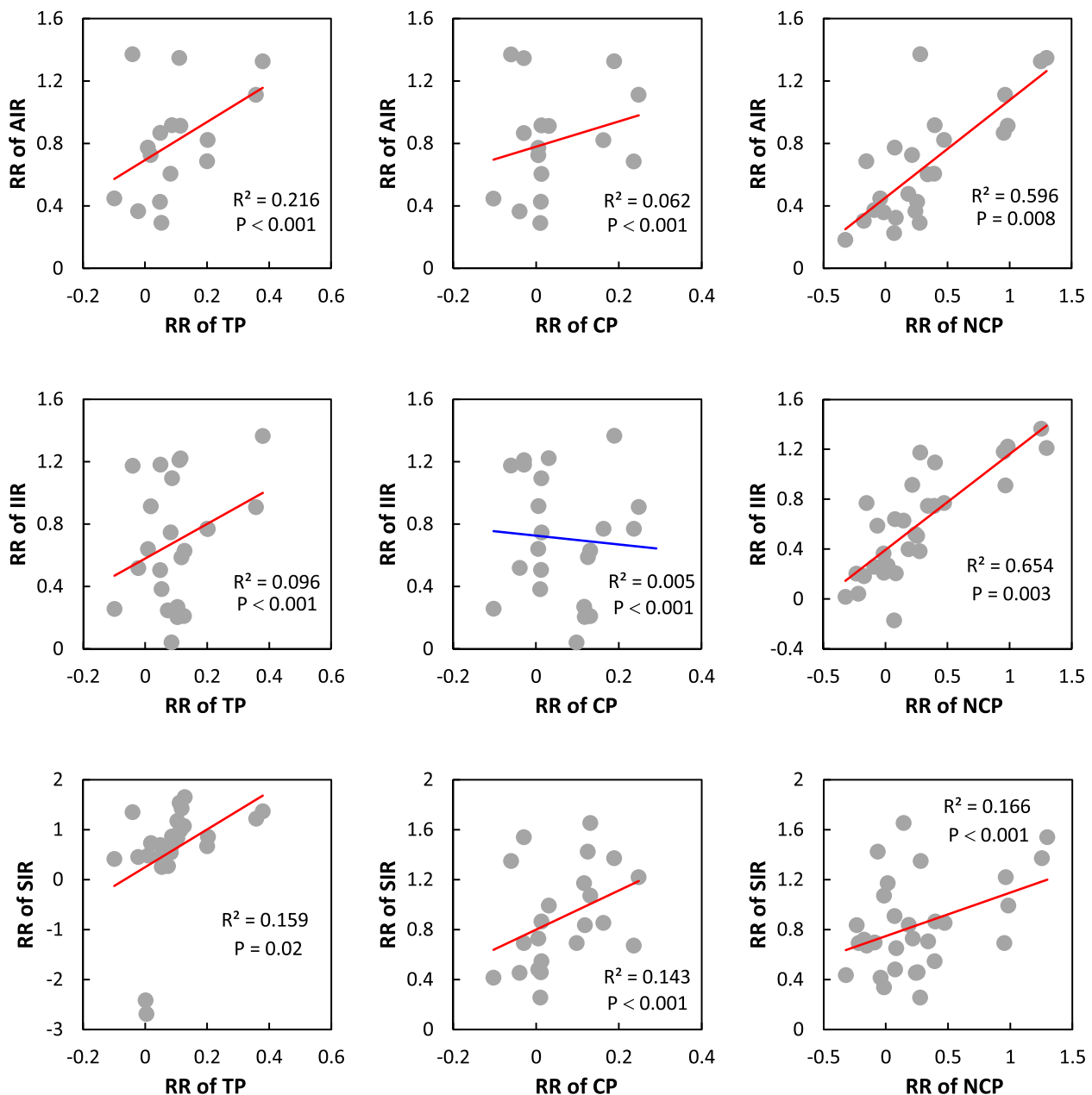


Fig. 5. Correlations between the RRs of total porosity (TP), capillary porosity (CP), and non-capillary porosity (NCP) and RRs of average infiltration rate (AIR), initial infiltration rate (IIR), and steady infiltration rate (SIR) (the solid red and blue lines indicate significant positive and negative correlations, respectively). RR represents the natural logarithm of the ratio between the experimental group and the control group.

growth (Pregitzer et al., 2000). Moreover, during the course of the year, fine roots tend to have a higher turnover rate than coarse roots (Archer et al., 2002), growing and decaying in response to seasonal changes in temperature and humidity (Pregitzer et al., 2000). Generally, fine roots exhibit growth from March to August, reaching peaks from June to August, after which growth rates decline and mortality gradually increases (Hansson et al., 2013; Konôpka et al., 2006; Mainiero et al., 2010; Xue et al., 2015). However, slight differences in the change patterns are observed among plant types. For example, growth of the fine roots of *Fraxinus mandshurica* tends to increase in spring compared to that in summer, and root death occurs primarily in spring and summer (Shi et al., 2007).

We also found that infiltration rates were influenced to a certain extent by soil depth, with increases being more evident in different soil layers compared to those observed in the soil profile as a whole, which presumably could be attributed to the differences in root growth in the

different soil layers. In this regard, root volumes are generally higher in the upper soil layers and gradually decrease with increasing soil depth (Archer et al., 2002; Yu et al., 2015). Moreover, during the course of development, root systems may undergo pronounced changes in certain soil layers, while showing less remarkable changes in other layers, which would contribute to the slight overall change when the entire soil profile is considered.

4.3. Implications and future prospects

The literature review performed for the present study indicated that there several studies have been conducted on the relationships between changes in root and soil hydraulic properties under certain soil and vegetation conditions. In these studies, researchers identified the correlations among root characteristics (e.g., root length density, root volume ratio, and root biomass) (Hao et al., 2020; Li et al., 2013; Liu

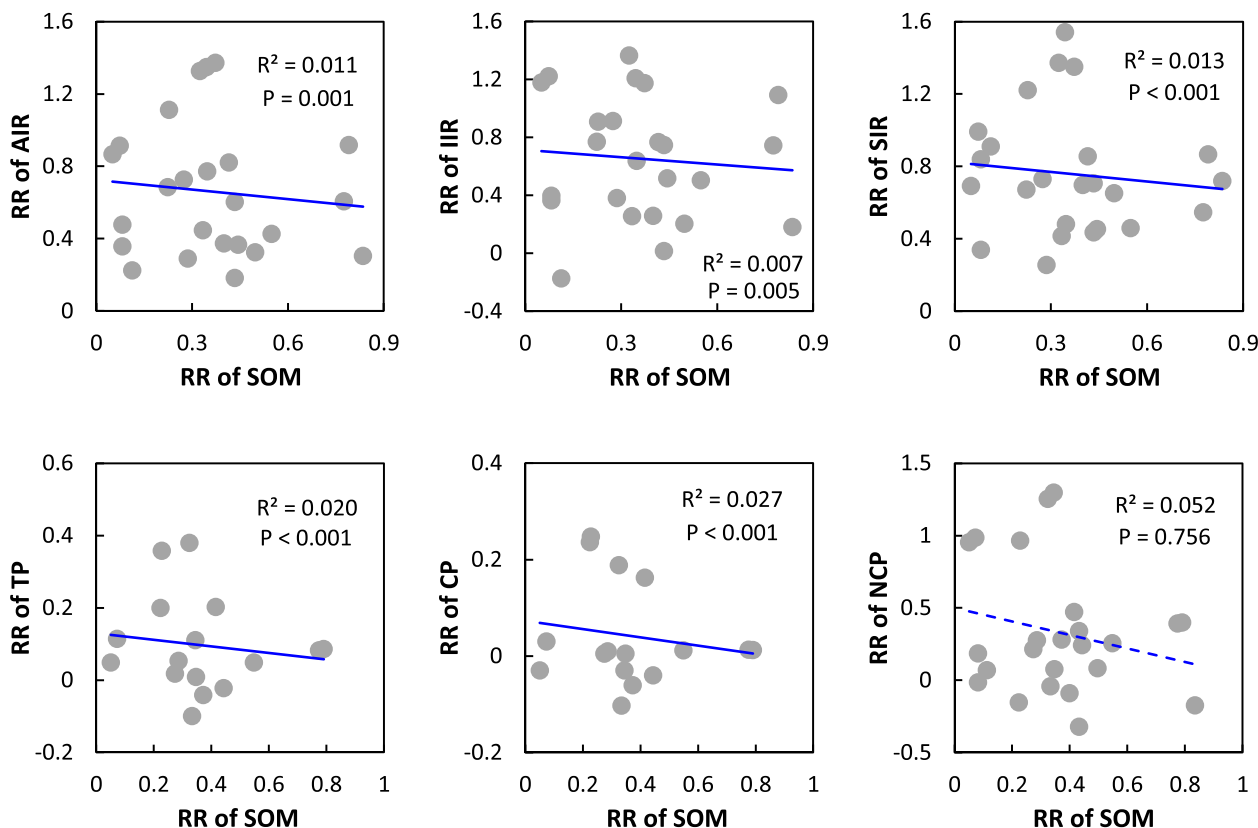


Fig. 6. Correlations between the RR of soil organic matter content (SOM) and the RRs of average infiltration rate (AIR), initial infiltration rate (IIR), steady infiltration rate (SIR), total porosity (TP), capillary porosity (CP), and non-capillary porosity (NCP) (solid blue lines indicate significant negative correlations, and the dotted line indicates that the correlation is not significant). RR represents the natural logarithm of the ratio between the experimental group and the control group.

et al., 2020; Wang et al., 2018; Yan et al., 2016), soil physical characteristics (e.g., porosity, organic matter content, and aggregates) (Cui et al., 2019; Jiang et al., 2018; Wu et al., 2016), and the corresponding soil hydraulic properties. Moreover, it has been established that most of the aforementioned variables show considerable correlations under certain conditions, and the corresponding relationships can be expressed by using equations. To determine whether there were similar correspondences between these characteristics under different conditions, we conducted a *meta-analysis* in which we integrated and analyzed the change characteristics of a range of relevant variables under different conditions and the correlations among these, with an aim of establishing general patterns characterizing the influence of root systems on soil hydraulic properties.

However, contrary to our expectations, our findings indicated that for most of the assessed interactions, the aforementioned variables showed notably less correlations under different conditions; consequently, we could not discern any clear patterns regarding the effects of root systems exerted on the hydraulic properties of soil under different conditions. Therefore, in future studies, it would perhaps be more feasible to focus on commonalities according to the corresponding relationships under certain specific conditions. For example, adjustment parameters may be set to facilitate the expression of general pattern equations, which will accordingly require strengthening of the experimental research methodology under specific conditions (i.e., examination of different permutations and combinations of different soil and vegetation conditions), and the establishment of datasets of root characteristics and the corresponding soil physical and hydraulic properties. In this regard, comparatively few studies have been conducted using continuous observation of plant roots in different seasons, and thus in the future, continuous observation experiments of different plants may be conducted to determine the influence of root development on soil

hydraulic properties under seasonal dynamics. Furthermore, most studies conducted to date have focused on the correlation between variables, whereas relatively few studies have assessed quantitative relationships. Thus, in future studies, substantial amounts of experimental data may be evaluated with an aim of toward establishing general equations for deducing the relationships between root characteristics and soil hydraulic characteristics under specific and general conditions. The information thus obtained may be used to refine existing soil water movement equations, which in turn may lead to improvements in regional or large-scale hydrological models and provide support for in-depth characterization of the terrestrial hydrological cycle. Moreover, this information may contribute toward the provision of guidance for agricultural and forestry management practices and engineering construction under similar conditions (Cannavo and Michel, 2013; Semwal et al., 2020; Wahren et al., 2009; Wang et al., 2020; Xie et al., 2020).

5. Conclusions

The development of root systems was explored to investigate the exertion of a significant influence on the hydraulic properties of soil, and based on a *meta-analysis*, we aimed to quantify this influence. Our findings revealed that root development could markedly enhance soil infiltration rates, saturated hydraulic conductivity, soil porosity, and soil organic matter content, particularly could notably promote increases in the rate of steady infiltration. However, it was observed that root-induced changes in soil porosity and organic matter content showed no correlation with changes in soil infiltration rates, with only non-capillary porosity exhibiting a moderate correlation with the observed changes in the initial and average infiltration rates. Moreover, among the assessed root characteristics, only root length density showed any notable (albeit low) correlation with changes in soil infiltration rate,

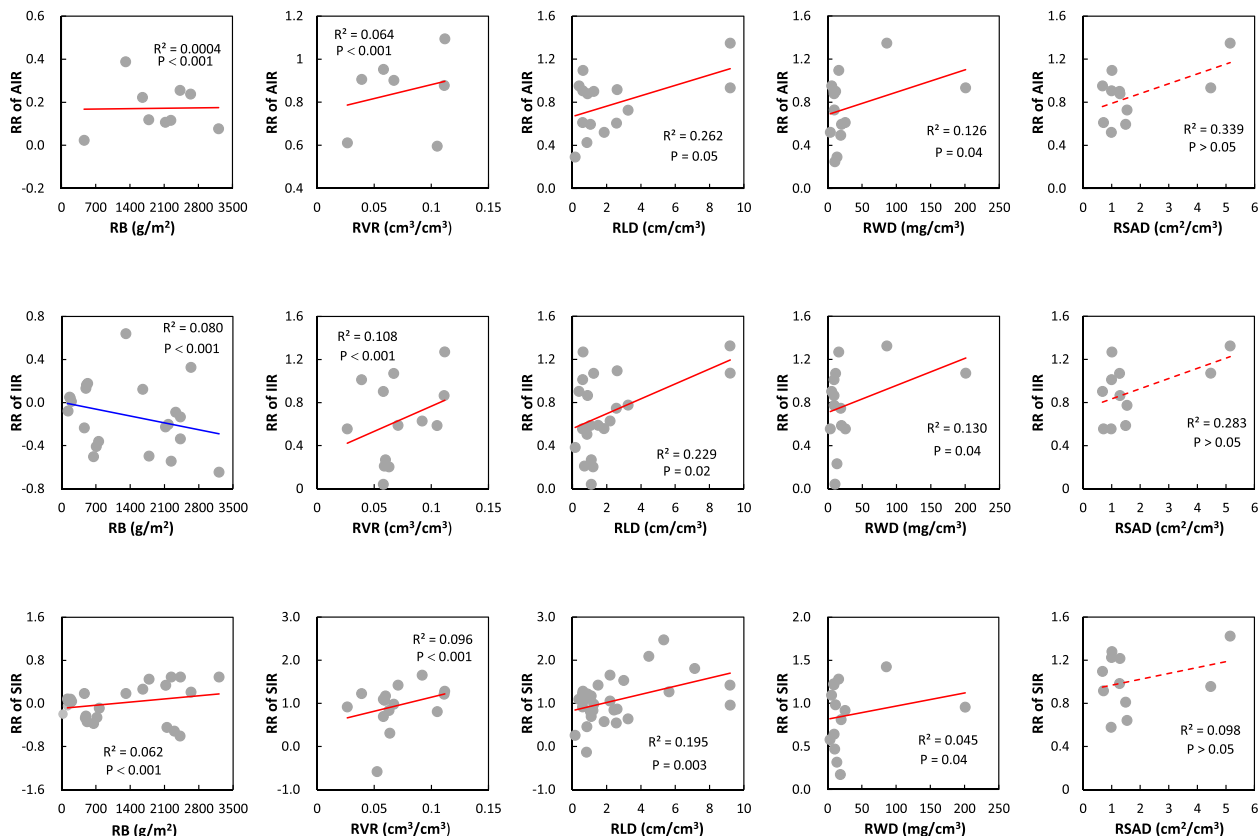


Fig. 7. Correlations between root characteristic variables [root biomass (RB), root volume ratio (RVR), root length density (RLD), root weight density (RWD), and root surface area density (RSAD)] and the RRs of average infiltration rate (AIR), initial infiltration rate (IIR), and steady infiltration rate (SIR) (the solid red and blue lines indicate significant positive and negative correlations, respectively, and the dotted lines indicate that the correlations are not significant). RR represents the natural logarithm of the ratio between the experimental group and the control group.

whereas other characteristics showed no correlation, which could be attributed primarily to the differing extents to which root systems exert influence on soil hydraulic properties under different conditions.

The influence of root development on soil properties was also found to be affected to varying degrees by soil texture, plant type, plant age, growing season, and soil depth. In this regard, our findings established that root systems exerted the most pronounced influence on infiltration rates and saturated hydraulic conductivity in loam soil, and that increases in soil hydraulic properties were found to be the highest in soils associated with older plants and in plants characterized by a greater number of tap roots. Furthermore, the influence of root systems on infiltration rates and saturated hydraulic conductivity exhibited significant seasonality, with increases being the most prominent in spring and summer and subsequently decreasing in autumn. Additionally, the effects of root systems on soil hydraulic properties differed in different soil layers. Considering this heterogeneity in the influence of root system development under different conditions, it will be necessary to conduct more comprehensive experimental studies under various conditions to determine the corresponding relationships among different variables. By doing so, it is anticipated that general patterns will emerge, which may provide a basis for conducting further in-depth studies on soil water movement and processes associated with the land phase of the hydrologic cycle, and may contribute toward project management.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2021.115363>.

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